

1.

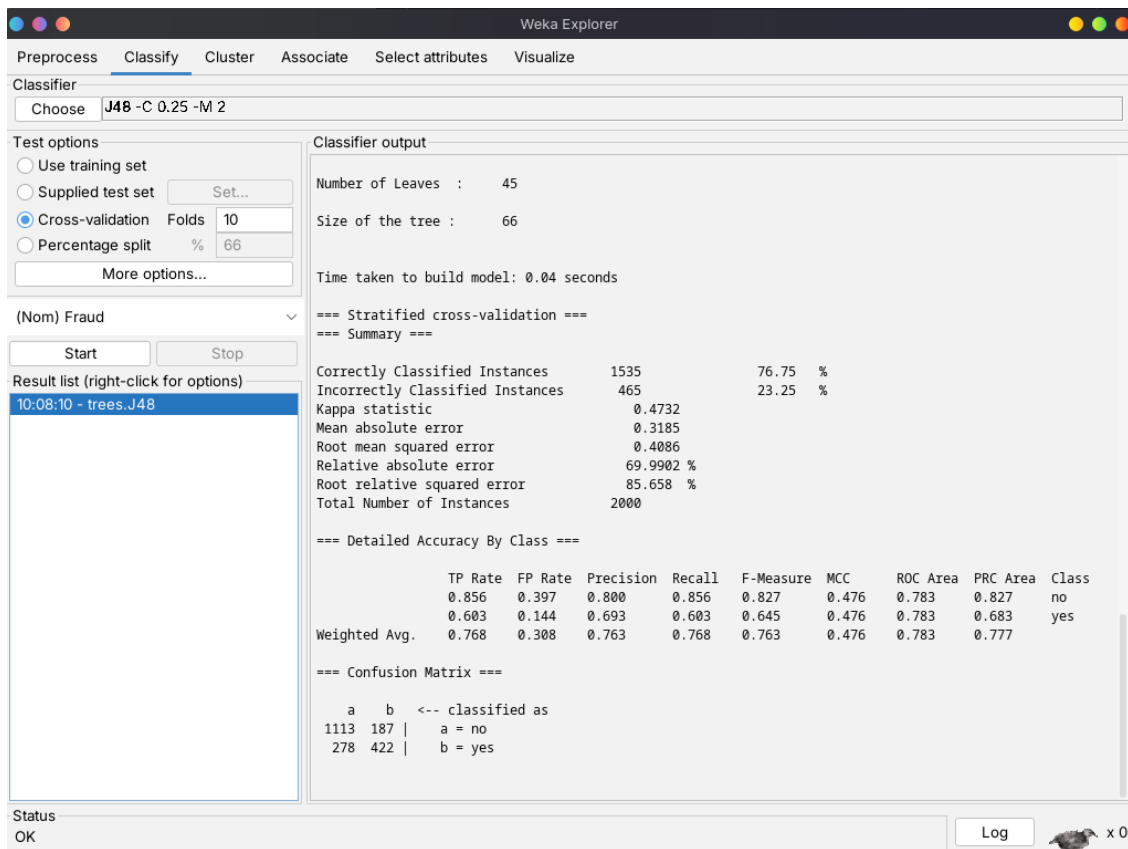


Figure 1.1: Tree based classifier

- The J48 Decision Tree classifier was trained using 10-fold cross-validation to ensure a balanced evaluation. Default parameters were used, and “Fraud” was selected as the class attribute.

Metric	Result
Correctly Classified Instances	1535 (76.75%)
Incorrectly Classified Instances	465 (23.25%)
Kappa Statistic	0.4732
ROC Area	0.783

- The J48 classifier achieved 76.75% accuracy, indicating a reasonably good performance. The Kappa statistic (0.4732) suggests moderate agreement beyond chance. The ROC area (0.783) implies good discrimination capability between fraudulent and non-fraudulent transactions.
- The confusion matrix shows that 1113 legitimate (“no”) and 422 fraudulent (“yes”) transactions were correctly classified. However, 278 fraudulent cases were missed, indicating room for improvement.

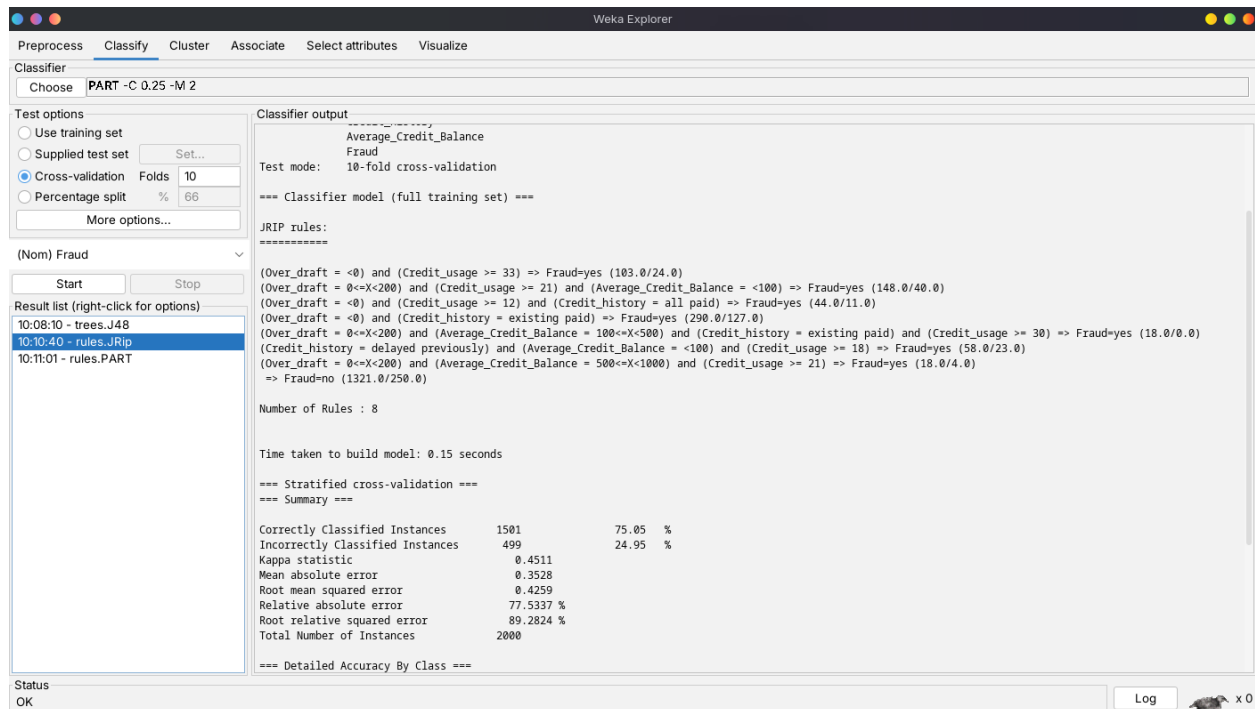


Figure 1.2: Rule set

- The JRip classifier achieved 75.05% accuracy, slightly lower than the J48 model. The Kappa value (0.45) again shows moderate agreement. The ROC Area (0.733) indicates good but not excellent classification ability.
- The following rules represent common fraudulent behavior identified by the model:
 - High credit usage (≥ 33) with overdraft < 0 --> Fraud likely.
 - Low average balance (< 100) and high usage (≥ 21) --> Fraud likely.
 - “All paid” credit history but with overdraft < 0 and moderate usage --> Fraud likely.
 - Existing paid credit history with small overdrafts --> May indicate subtle fraudulent cases.
- These rules highlight that negative overdraft and excessive credit utilization are strong fraud indicators.
- JRip’s rule-based output provides better interpretability than J48, making it useful for financial analysts.

2.

$H_0 = 0.150, H_1 = 0.090, H_2 = 0.120, H_3 = 0.065, H_4 = 0.175$

$A_0 = 0.290, A_1 = 0.060, A_2 = 0.180, A_3 = 0.150, A_4 = 0.240$

$S < 0.15$ is considered as normal

$$s = \frac{1}{2} \sum_{i=0}^4 |A_i - H_i|$$

- $|A_0 - H_0| = |0.290 - 0.150| = 0.140$
- $|A_1 - H_1| = |0.060 - 0.090| = 0.030$
- $|A_2 - H_2| = |0.180 - 0.120| = 0.060$
- $|A_3 - H_3| = |0.150 - 0.065| = 0.085$
- $|A_4 - H_4| = |0.240 - 0.175| = 0.065$
- $S = \frac{1}{2} \times (0.140 + 0.030 + 0.060 + 0.085 + 0.065) = 0.190$
- $0.190 > 0.15$, the profile is not normal
- Because S exceeds the threshold by a non-trivial margin (0.190 vs 0.150) Donald's recent access behavior as anomalous.
- The largest contributors are indices 0 and 3 (A_0 rose from 0.150 to 0.290 and A_3 from 0.065 to 0.150). so the anomaly is driven mainly by those file-operation types. multiple other indices also increased suggesting a systemic change. This warrants investigation of the processes, timestamps and files involved for indices 0 and 3.

3. Part A

- $C = 103982696638$
- Public key: $(n, g) = (426749, 709)$
- Private key: $\lambda = 212718, \mu = 2748$
- $n^2 = 426749^2 = 182114709001$
- $u = C^\lambda \bmod n^2$

$$103982696638^{212718} \bmod 182114709001 = 36324874881$$

$$L(u) = \frac{u-1}{n}$$

$$L(u) = \frac{36324874880}{426749} = 85120$$

$$m = (L(u) \cdot \mu) \bmod n$$

$$m = (85120 \times 2748) \bmod 426749$$

$$m = 51308$$

Part B

- Key generation: Alice, acting as the receiver, first creates a public and private key pair using the Exponential ElGamal cryptosystem. She keeps the private key secret and sends the public key to both of her shops.
- Encryption at the shops: Each shop encrypts the number of phones sold using Alice's public key and a random number, so the actual quantity remains hidden. They then send this encrypted value along with the price rate to the cloud server.
- Computation at the cloud: The cloud uses the homomorphic property of Exponential ElGamal to compute the total earning without decrypting anything. It raises each encrypted number to the power of its price rate and multiplies the results together to produce an encrypted total earning.
- Decryption by Alice: The cloud sends this final encrypted total to Alice, who decrypts it using her private key to find the actual total earnings.

4.

- Yes, even a good secret-keeping method (cryptography) can eventually be broken if a hacker uses super-powerful computers like specialized gear or giant cloud systems designed for guessing secrets.
- Security steps:
 - Use Newer, Tougher Secret Methods
 - Instead of older RSA, start using Elliptic Curve Cryptography (ECC). It's secure but uses much shorter keys, which is faster.
 - For the main message encryption, use AES-256 in GCM mode (AES-GCM). This is the top-tier method because it not only scrambles the message but also makes sure no one has secretly messed with it.
- Combine and Secure the Process
 - Use a hybrid lock system. Encrypt the big message quickly using AES-GCM with a temporary key. Encrypt just that temporary key using public-key method (like RSA or ECC).
 - Use temporary keys (like ECDHE) for each communication session. This means if a hacker steals your main, long-term secret key later on, they still can't unlock your past messages.
- Keep Secrets Safe and Random
 - Make sure you create keys using truly random sources, avoiding predictable patterns.
 - Store the secret keys in super-safe places, like special chips or hardware modules designed just for that.
- Layered Defenses
 - Set limits on how many times someone can try to log in (rate-limiting) and constantly watch for unusual hacker patterns (monitoring).
 - Use intrusion detection systems to log everything and give an immediate warning if an attack is happening.
- Plan for Quantum Attacks
 - Since super-powerful quantum computers might eventually break today's ECC and RSA, start planning now to switch to post-quantum encryption methods that are safe from those future computers.

References

- CalculatorSoup, L. (2025). *Large Exponents Calculator*. [online] CalculatorSoup. Available at: <https://www.calculatorsoup.com/calculators/algebra/large-exponent-calculator.php> [Accessed 5 Nov. 2025].
- Calculator.net. (2025). *Big Number Calculator*. [online] Available at: <https://www.calculator.net/big-number-calculator.html?cx=36324874880&cy=426749&cp=20&co=divide> [Accessed 5 Nov. 2025].
- Sandiego.edu. (2025). *Power Mod Calculator*. [online] Available at: https://home.sandiego.edu/~cparker/old_classes/math370_sp18/powermod.html [Accessed 5 Nov. 2025].